

Lab: Keyspace Operations



Objectives

After completing this lab, you will be able to:

- Create, update, and remove a keyspace in Cassandra
- Identify the keyspace for the current session with the USE command
- Describe a keyspace by listing all the tables in it

Exercise 1 - Getting the environment ready

Open the Cassandra database page by clicking the button below:

[Open Cassandra Page in IDE](#)

On that page, click the [Create](#) button to create a Cassandra database.

After creation has finished, click the [Cassandra CLI](#) button (below the [Create](#) button from the previous step) to connect to the database using the cqlsh CLI.

[←](#) [→](#)

DATABASES

BIG DATA

CLOUD

EMBEDDABLE AI

OTHER

Launch Application

Cassandra

ACTIVE

v4.1 | v5.0.1

Connect to Cassandra directly in your Skills Network Labs environment.

CreateDelete

SummaryConnection InformationDetails

Your database server is now ready to use and available with the following login credentials. For more details on how to navigate Cassandra, please check out the Details section.

Username:ns1mben1aya

Password:OTA4NC1uYm1tYWVv

To interact with the database in the terminal, select one of these options:

Cassandra CLINew Terminal

```
er-na1mben1aya /home/project theia@theiadocker-na1mben1aya: /home/project theia@theiadocker: |  
theia@theiadocker-na1mben1aya:/home/project$ cqlsh 172.21.215.234 9042 --username r  
oot --password jCxm283gkks7H8DgMAPJcNu1  
WARNING: cqlsh was built against 4.0.17, but this server is 5.0. All features may n  
ot work!  
Connected to Test Cluster at 172.21.215.234:9042  
[cqlsh 6.0.0 | Cassandra 5.0-beta1 | CQL spec 3.4.7 | Native protocol v5]  
Use HELP for help.  
root@cqlsh>  
root@cqlsh> |
```

Exercise 2 - Create a keyspace

The below command creates a keyspace called **training**, using SimpleStrategy and a replication_factor of 3.

SimpleStrategy is used when all the nodes in your cassandra cluster exist in a single data center.

On cqlsh run the below command.

```
1 CREATE KEYSPACE training
2 WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 3};
```

List all keyspaces.

```
1 describe keyspaces
```

Exercise 3 - Describe a keyspace

In the previous exercise you have created a keyspace named **training**.

Let us print more details of it using the describe command.

Describe a keyspace.

```
1 describe training
```

```
root@cqlsh> CREATE KEYSPACE training
... WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 3};

Warnings :
Your replication factor 3 for keyspace training is higher than the number of nodes 1

root@cqlsh> describe keyspaces

system      system_distributed  system_traces  system_virtual_schema
system_auth  system_schema       system_views    training

root@cqlsh> describe training

CREATE KEYSPACE training WITH replication = {'class': 'SimpleStrategy', 'replication_factor': '3'} AND durable_writes = true;
root@cqlsh>
```

Exercise 4 - Alter a keyspace

In a previous exercise, you created a keyspace named **training** using SimpleStrategy.

Let's change that to use NetworkTopologyStrategy.

NetworkTopologyStrategy is used when all the nodes in your cassandra cluster are spread across multiple data centers.

Alter a keyspace.

```
1 ALTER KEYSPACE training
2 WITH replication = {'class': 'NetworkTopologyStrategy'};
```

Verify the changes using the describe command.

```
1 describe training
```

```
root@cqlsh> ALTER KEYSPACE training
... WITH replication = {'class': 'NetworkTopologyStrategy'};

Warnings :
Your replication factor 3 for keyspace training is higher than the number of nodes 1
for datacenter datacenter1

root@cqlsh> describe training

CREATE KEYSPACE training WITH replication = {'class': 'NetworkTopologyStrategy', 'datacenter1': '3'} AND durable_writes = true;
root@cqlsh>
```

Exercise 5 - Use a keyspace

To use a keyspace run the below command.

```
1 use training;
```

List all tables in this keyspace.

```
1 describe tables
```

You will get an empty list because we do not have any tables yet.

Exercise 6 - Drop a keyspace

To drop a keyspace, run the below command.

```
1 drop keyspace training;
```

Verify the changes using the describe command.

```
1 use system;
2 describe keyspaces
```

Practice exercises

1. Problem:

Create a keyspace named sales using SimpleStrategy and a replication factor of 1.

▼ Click here for Hint

use the 'create keyspace' command

▼ Click here for Solution

On the cqlsh run the below command.

```
1 CREATE KEYSPACE sales
2 WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 1};
```

2. Problem:

Change the replication factor to 3 for the sales keyspace.

► Click here for Hint

▼ Click here for Solution

On the cqlsh run the below command.

```
1 ALTER KEYSPACE sales
2 WITH replication = {'class': 'SimpleStrategy', 'replication_factor' : 3};
```

3. Problem:

Drop the sales keyspace.

► Click here for Hint

▼ Click here for Solution

On the cqlsh run the below command.

```
1 drop keyspace sales;
```

4. Problem:

Verify that the sales keyspace is dropped.

► Click here for Hint

▼ Click here for Solution

On the cqlsh run the below command.

```
1 describe keyspaces
```

5. Problem:

Disconnect from the cassandra server.

▼ Click here for Hint

Refer to the [Using the CQL Shell \(cqlsh\) lab](#).

▼ Click here for Solution

Run the below command on the terminal.

1

`exit`

